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The flipped classroom: A meta-analysis of effects on student performance across disciplines and education levels

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ABSTRACT

The Flipped Classroom design has arguably revolutionized teaching across the globe. In this article we report the first comprehensive meta-analysis of its effects on student performance, relative to traditional teaching models, across disciplines and education level. We included 198 studies comprising 33,678 students in our meta-analysis. There were 174 studies conducted at the tertiary level, 21 at secondary, and three at primary school level. Overall, the flipped classroom had a moderate positive effect (g = .50) on student performance. The flipped classroom was beneficial regardless of discipline, with effect sizes ranging from weak (i.e., for IT, $g = 0.30$; $k = 14$) to strong (i.e., for humanities, $g = 0.98$; $k = 34$). Testing of other moderators relating to group sizes, in-class activities, pre-class testing, and the sophistication of a flipped classroom design provide evidence to suggest that the primary contributing factor to the flipped classroom effect is the opportunity it provides for structured, active learning and problem-solving.

1. Introduction

The flipped classroom has become a popular and strongly advocated approach to student learning in the K-12 sector as well as in higher education. Recent Horizon reports (New Media Consortium, 2014, 2015) identified flipped classrooms as important developments in educational technology, and the 2018 report (Becker et al., 2018) noted the changing role of academics and highlighted that academics could be expected to work with a variety of teaching models including flipped classes. A recent search of Google Scholar (August 2019) yielded approximately 70,000 and 60,000 hits for the terms 'flipped classroom' and 'inverted classroom' respectively. However, does flipping the classroom actually improve student performance? If so, by how much, and in what disciplines and at what level of student learning is it most effective? Moreover, can we identify the crucial ingredients that contribute to the flipped learning effect, presuming there is one?

There have been studies of various kinds examining the impact of the flipped classroom, including systematic reviews (e.g., Betihavas, Bridgman, Kornhaber, & Cross, 2016; Chen, Lui, & Martinelli, 2017; Lundin, Bergviken Rensfeldt, Hillman, Lantz-Andersson, & Peterson, 2018), scoping reviews (e.g., O'Flaherty & Phillips, 2015; Tang et al., 2018) and literature reviews (e.g., DeLozier & Rhodes, 2017; Velegol, Zappe, & Mahoney, 2015; Ward, Knowlton, & Laney, 2018). We are also aware of five peer-reviewed published meta-analyses, but these have been small and local, focusing on effects within specific disciplines, that is, health professions education (es = 0.39; 28 studies; Hew & Lo, 2018), mathematics (es = 0.30; 21 studies; Lo, Hew, & Chen, 2017), pharmacy (no significant effect; six studies; Gillette et al., 2018), nursing knowledge (es = 1.06) and skills (es = 1.40; 12 studies; Hu et al., 2018), and various disciplines in the Korean education setting (es = 0.47; 36 studies; Cho & Lee, 2018).

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Generally speaking, the previous reviews suggest that flipped classrooms tend to have a positive effect on student performance. However, a study looking at overarching effects of the flipped classroom in a quantitative manner is required. In this paper we therefore report a comprehensive meta-analysis of the effects of the flipped classroom model across disciplines and education level based on an initial pool of over 6000 articles. We also analyse the extent to which different aspects of flipped learning moderate its effects on student performance.

1.1. The flipped classroom model

According to one prominent definition, "the flipped approach inverts the traditional classroom model by introducing course concepts before class, allowing educators to use class time to guide each student through active, practical, innovative applications of the course principles" (Academy of Active Learning Arts and Sciences, 2018). In another well-recognized conceptualization, the flipped classroom reflects a "set of pedagogical approaches that (1) move most information-transmission teaching out of class; (2) use class time for learning activities that are active and social; and (3) require students to complete pre- and/or post-class activities to fully benefit from in-class work" (Abeysekera & Dawson, 2015, p. 3).

Why would teachers and course coordinators flip their classes and courses? Arguably the most parsimonious answer is that educators have come to realize that a flipped approach provides for the most productive use of their face-to-face time with students, a pedagogical point upon which we elaborate shortly (for overviews of the historical roots and benefits of the flipped classroom model see Bergmann & Sams, 2014, and Talbert, 2017; for some notable precursors to the flipped classroom see peer learning (Crouch & Mazur, 2001) and just in time teaching (Novak, Patterson, Gavrin, & Christian, 1999)). But to some extent, educators have also been motivated to meet changing demands in the education marketplace. At universities, for example, lecture attendance has been decreasing over several years, due to students' changing work habits, increased desire for flexible education delivery, and because the Internet now makes gathering high quality content trivial for many students. Advocates of the flipped classroom suggest that the flipped model may help educators better meet these students' needs (e.g., Clark, 2015). At the secondary school level, for example, technological advances now make it possible for teachers to quickly produce informal but informative presentations and post them online, thereby enabling students to learn in their own time and save teachers the unnecessary effort of having to repeat themselves to these students (e.g., Bergmann & Sams, 2014).

At the same time, there may be barriers to the efficacious implementation of a flipped classroom. Because it is a challenge to traditional approaches to teaching, successful flipped classrooms require motivated and confident teachers, who also require appropriate levels of time, resources, and support to develop a flipped model (for a brief review, see Wang, 2017). Students themselves may be resistant to the flipped model, since it requires more active learning on their part, and greater pre-class preparation (e.g., Sander, Stevenson, King, & Coates, 2000).

In addition to pragmatic imperatives, the flipped classroom model has been developed on the bases of theory and empirical research. For example, at the tertiary level, traditional lectures have tended to be didactic so that students are passive receptacles of information, and the information presented in lecture theatres is usually new; thus, students' learning is 'surface' rather than 'deep' (e.g., Beattie, Collins, & McInnes, 1997). In addition, there is evidence that students retain less than 50% of information presented during a traditional 50-min lecture (for a review see Wilson & Horn, 2007). To address these realities, students in the flipped classroom model are typically introduced to new material online via presentations that are short, ideally five-ten minutes long, reflecting typical student attention spans (e.g., Hartley & Cameron, 1967) and are required to engage with this material prior to attending a class with their teacher. Further, students who can pause and rewind online presentations—as opposed to frantically taking notes in a lecture theatre—may be better able to better manage cognitive load and retain more information with greater depth (Clark, Nguyen, & Sweller, 2005).

It is here, in the classroom or tutorial or even a lecture theatre setting, that the flipped classroom approach could be beneficial in terms of enhancing student learning. Students work on applying their new-found knowledge by engaging in interactive tasks, exercises, and activities—usually requiring them to solve problems within a small group—with the teacher on hand to provide guidance and trouble-shooting advice. There is evidence to support the efficacy of both problem-based learning approaches (Hmelo-Silver, 2004) as well as small group learning (for a meta-analysis, see Springer, Stanne, & Donovan, 1999). When combined, these types of learning sessions appear to possess the ingredients that encourage active learning (e.g., Prince, 2004), which refers to both 'doing'—rather than simply reading or hearing about a concept—and thinking about what one is doing when engaging in a learning task (Bonwell & Eison, 1991). Meta-analyses comparing active and passive learning approaches indicates that active learning is more likely to be associated with improved student performance (Freeman et al., 2014; Richardson, Abraham, & Bond, 2012).

In addition, whereas traditional teaching approaches tend to disregard fundamental student needs of autonomy, competence, and relatedness, active learning approaches are likely to facilitate them, in turn encouraging higher levels of intrinsic student motivation to learn (Abeysekera & Dawson, 2015). The flipped classroom model also recognises the importance of repetition for enhancing retrieval when learning (e.g., Karpicke & Roediger, 2008), insofar as the work undertaken in class reflects application of new information received before class. Accordingly, students often do some form of assessment—usually formative—following completion of a pre-class activity such as a multiple-choice test relating to a pre-recorded lecture presentation. This provides them with an indication of what they know, and a baseline for tracking improvement in relation to summative assessment that often follows the in-class interactive activities, whether immediately and/or at the completion of the course or learning unit.

1.2. Salient variables when comparing flipped and traditional teaching approaches

There are many inter-related components in a flipped classroom model. Therefore, in our meta-analysis, we sought to take account of the variables that the literature suggested may impact on the extent to which flipped (versus traditional) approaches to teaching affect student performance. These are moderating variables, which we assessed in addition to the type of study design (within or between groups) and information relating to student level (primary, secondary, tertiary) and discipline.

We were cogniscent of the fact that there are many 'versions' of the flipped classroom (for a discussion, see Talbert, 2017). For example, some teachers consider assessment to be an integral part of the process, whereas others do not. For some, flipped classrooms are defined by pre-class technology, whereas for others, it is not important how pre-class material is delivered or experienced. And sometimes teachers claim to employ a flipped model, when in fact all they have done, for example, is put their lecture recordings online but not provided opportunity for in-class interaction. Thus, after studies passed our screening test, we coded them according to whether a classroom was 'classically' flipped according to our criteria (specifically, pre-class preparation of any type, some form of pre-class engagement measure, and student activities in class with or without in-class lecture) or 'minimally' flipped (i.e., the study *did not* incorporate one or two or all three of pre-class preparation, a measure of pre-class engagement, or in-class activities). In addition, we coded studies according to whether the entire course or only a disparate element (i.e., a topic, unit, module) was flipped. We also coded whether the instructor was the same across the flipped and traditional classes, to assess the extent to which teacher quality may affect student performance across flipped and traditional approaches (Darling-Hammond, 2000).

There is some evidence that pre-class learning can influence post-class outcomes (e.g., Moravec, Williams, Aguilar-Roca, & O'Dowd, 2010). Thus, we coded studies according to whether there were measures of pre-class engagement and, if so, what form they took (specifically, formative or summative assessment; discussion at beginning of in-class activities; reflective learning questions/activities).

The nature of the in-class activities in a flipped classroom may play an important role in the extent to which the flipped classroom improves student performance (for a discussion of in-class activities, see Bergmann & Sams, 2014; Talbert, 2017). For example, a course may technically be flipped, insofar as a lecturer posts material online for students to watch prior to attending class—but students' in-class engagement may still be passive, rather than active, if the in-class activity still consists of a didactic lecture. Alternatively, there may be differences in student performance depending on whether in-class activities are predominantly student-centred or teacher-guided. Thus, we coded flipped in-class activities according to whether they were standard didactic (lecture only); predominantly teacher-guided (e.g., a discussion or Q&A session on pre-class work, with/without a lecture, but no student in-class activities); predominantly student-centred, whereby in-class activities are a feature, and the majority of the work is driven or generated by students, for example, engaging in problem-solving tasks (but could also involve teacher discussions, Q&A sessions, mini-lectures as per above); or not reported.

Given that small group work is a feature of the in-class component of the flipped classroom, it was possible that group size could affect the efficacy of a flipped classroom approach. Thus, we coded studies according to the size of the small groups in which students worked. We selected group sizes based on commonly reported effective groups. Some scholars argue that the ideal small group size in a learning context is three or four (e.g., Oakley, Felder, Brent, & Elhajj, 2004). Others (e.g., AbuSeileek, 2012) report that a group size of five outperforms other group sizes in communication tasks. Still other studies suggest that increasing group size from pairs to four shows little difference (Pfister & Oehl, 2009). In work teams, six to nine are described as the best performing size (Williams, Champion, & Hall, 2011); others suggest 10 (e.g., Kleiner, 2014). We therefore coded for group size around these boundaries, specifically, 5 students or less; 6–10 students; > 11 students; described as 'small group' but number not quantified; described as 'group' but number not quantified; and not reported.

Finally, we explored whether the nature of the student summative assessment task could influence results—specifically, the extent to which assessment that was more robust may indicate different outcomes compared to assessment that arguably was less so. We defined a robust assessment as having *at least* 20 MCQs or six short answer questions. We based our criteria on the notion of reliability of the assessment as a whole. Although 20 MCQs does not guarantee a high reliability, literature does recommend initial sets of 5–30 (e.g., Considine, Botti, & Thomas, 2005). Erring on the longer side we settled on 20 MCQs. Working on the basis that students take 1 min per MCQ, this equated to approximately six short answer questions. Thus, we coded robust assessment as any assessment containing > 20 MCQ, or > 6 short answer questions, or an essay, or a skills-based test, or an exam mark, or an overall course mark. We coded less robust assessment as < 20 MCQs, or true/false items, or fill-in-the-blank items, or matching items; or writing a paragraph, or < 6 short answer questions.

1.3. Research questions

Our primary research question was, to what extent is a flipped classroom model associated with better student performance relative to traditional approaches to teaching?

Based on the preceding discussion, we asked the following additional research questions, corresponding to the many moderating variables that we tested.

1./Does the flipped classroom effect differ according to whether students are primary, secondary, or tertiary?
2./Does the flipped classroom effect differ across discipline?
3./Does it matter if the classroom is 'classically' flipped or 'minimally' flipped? In other words, when it comes to student performance, is it necessary that all the ingredients that are generally recognized as being essential to a flipped classroom be present

when the classroom is flipped?
4./Relatedly, in terms of student performance, does it matter if an entire course or only disparate elements within a course (e.g., a topic, unit, or module) are flipped?
5./To what extent is student performance affected if the instructor is the same (or different) across both the flipped and traditional classes?
6./To what extent does the form of pre-class engagement affect student performance?
7./To what extent does the type of in-class activities in a flipped classroom affect student performance?
8./Does group size affect student performance?
9./To what extent does the nature of the student assessment task—specifically, robust versus less robust tasks—affect student performance?
10./Does the type of study design (within or between groups) affect student performance?

2. Method

2.1. Procedures

2.1.1. Literature search and inclusion criteria

We employed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA; Moher, Liberati, Tetzlaff, & Altman, 2009) guidelines in the design and reporting of this meta-analysis. We conducted a comprehensive search of the research literature, published prior to January 2018, using the Pubmed, Scopus, PsycINFO, Academic Search Complete, Education Research Complete, ERIC and Teacher Reference Center electronic databases under the guidance of an expert research librarian (see Supplementary Table 1, for detailed search strategies). In addition, the reference lists of included articles were checked for other relevant papers.

Each study had to meet the following criteria in order to be included in this meta-analysis:

1) it examined performance outcomes for both a flipped/inverted group (as described by study authors in either the title or abstract) and a traditional didactic (lecture) group;
2) the study design was either
   a. between-groups: a flipped and traditional course run concurrently or a flipped course compared with the same course taught in a traditional format in previous semesters or
   b. the same students participating in both flipped and traditional components of a course (within-group); and
   c. data were provided in a format that enabled calculation of an effect size for differences between the performance of flipped and traditional groups;
3) it was published in a peer-reviewed journal ('gray' literature was excluded) in English, and
4) it was original research with $N > 1$ participant (excludes reviews & case-studies).

The literature search initially identified 6166 articles, 857 of which were duplicates. The titles and abstracts of the remaining 5309 articles were screened by the second author (AJO), after which full-text versions of 739 studies were retrieved for detailed screening. Re-application of the inclusion criteria reduced the number of eligible studies to 207 (see Fig. 1, for details of the review & selection process). However, all participants were required to be independent of those used in other eligible studies because the magnitude of an effect size may be distorted by the inclusion of non-independent samples (Rosenthal, 1995). Thus, all studies were checked to establish independence. The final number of eligible studies was 198, with a combined total of 33,678 students (see Supplementary Table 2 for summary details of each study).

2.1.2. Data collection and preparation

Educational details (level, subject, type of assessment), characteristics of the flipped course (instructor, measure of pre-class engagement, type of in-class activity, size of in-class groups, extent of flip, quality of student assessment), study design (between, within) and data for performance outcomes were extracted from each study (see Supplementary Table 2 for summary details). Two independent raters coded variables on all 198 papers, with 90% inter-rater agreement. Where there were discrepancies between raters, papers were reviewed again and, if the specific methodology was still not clear, a consensus decision was reached by all three authors.

Some simple transformations were needed to standardise the data before it could be analysed. First, standard errors (SE) were transformed into SDs where needed (Hedges, 1982). Second, where only the median, range or interquartile range were provided, means and standard deviations were estimated using the methods recommended by Wan, Wang, Liu, & Tong, 2014. When data were reported for flipped/traditional subgroups that were not relevant to the current analysis (e.g., males vs females), the data were combined in order to generate a single, overall effect size for inclusion in the current meta-analysis.

Where a study did not report the sample size of one of their groups (e.g., the traditional group from an earlier year), but all other necessary data were available, study authors were contacted and asked to provide sample size information. Of the 13 authors contacted, four subsequently reported these data, enabling these studies to be included (Clark, 2015; Girgis & Miller, 2018; Haughton & Kelly, 2015; Pierce & Fox, 2012).

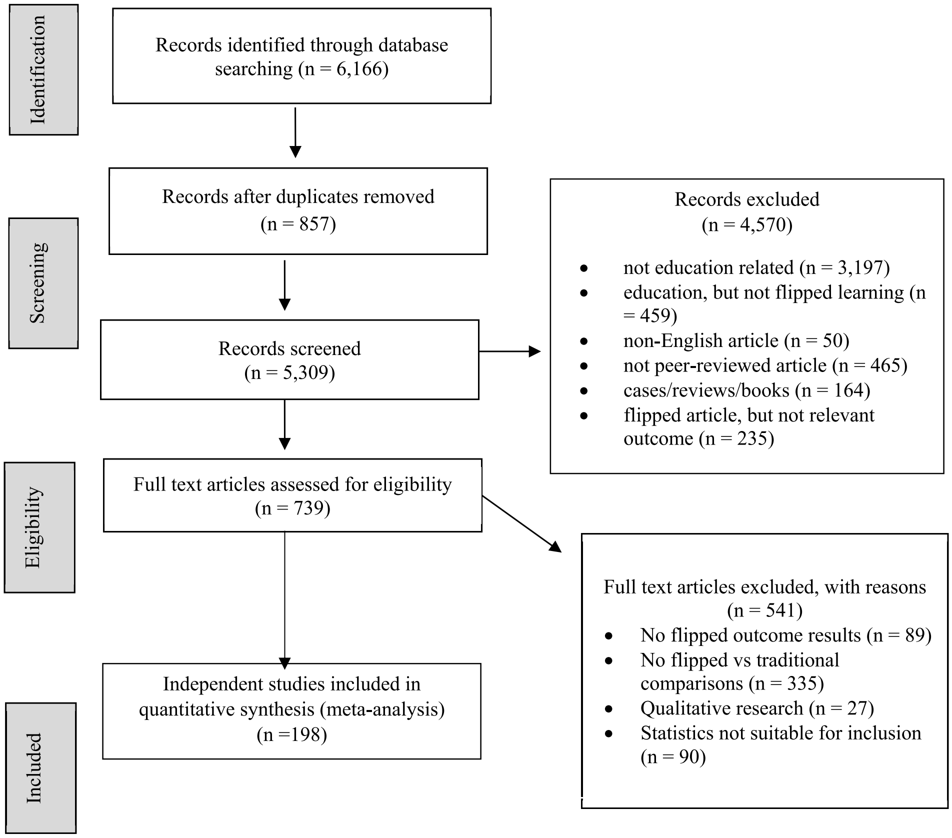


Fig. 1. Flowchart of study selection process.

2.2. Statistical analyses and interpretation

We performed all analyses using the Comprehensive Meta-Analysis software (CMA, version 3; ©2014, Biostat, Inc., Englewood, NJ). The raw (outcome) data were most frequently provided in the form of means and SDs, from which Hedges' g effect sizes were calculated in order to measure the standardised mean difference between the outcomes of the flipped and traditional groups. Data were additionally provided in the form of t-values, standardised mean differences, and exact p values, which, in conjunction with sample sizes, enabled Hedges' g to be calculated in CMA.

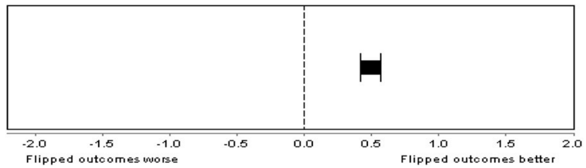
Mean effect sizes were calculated by weighting individual effect sizes by their inverse variance and them pooling them to create a mean weighted effect (g_w). This weighting considers the fact that the reliability of an individual effect is affected by the size of the sample from which it is derived: effect sizes calculated from larger samples are more precise, have a smaller variance and are, therefore, given a higher weighting (Hedges, 1985; Lipsey & Wilson, 2001). A positive g_w indicates that the flipped learning group achieved better performance outcomes than the traditional group. Similarly, a negative g_w indicates that those being taught in a flipped learning environment performed more poorly than their peers in the traditional group. Effect sizes were interpreted using Cohen's guidelines of g_w = .2, .5 and 0.8 equating to small, medium and large effects, respectively (Cohen, 1992).

Given the findings from different studies were heterogeneous, a conservative random-effects model was used to calculate mean effect sizes. This model assumes that the effect sizes for individual studies vary as a result of sampling error and study design, and contrasts with that of a fixed-effects model, which assumes that sampling error is the only source of variability (Borenstein, Hedges, & Higgins, 2009). Importantly, where a study reported multiple scores (e.g., both student exam scores and course grades) and these outcomes were eligible to be included in the same analysis, a mean effect was calculated to ensure that each study contributed only one effect size to any given analysis (Lipsey & Wilson, 2001). Studies were also grouped to examine the potential impact of moderating variables (e.g., type of assessment) because it is suggested that the variability in the findings of different studies should be explored via sub-group analyses using random-effects models (Borenstein et al., 2009; Stroup et al., 2000).

Forest plots were generated using Forest Plot Viewer (Boyles, Harris, Rooney, & Thayer, 2011) in order to examine effect size distributions and to assist in identifying outliers, and 95% confidence intervals (95% CIs) were calculated to provide a measure of the precision of the mean effect size estimate and are used to determine statistical significance (i.e., if the 95% CIs do not include zero, this indicates that there is a significant difference between the mean performance outcomes of the flipped and traditional groups). Probability (p) values were additionally calculated to assess the statistical significance of the effect sizes.

Heterogeneity was assessed using the Q-statistic, with significant values indicating variability in the effect sizes reported by

Table 1
Summary of effect sizes for all studies combined, education level, and subject discipline.

a.	Overall										
	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' g	Lower CI	Upper CI	p	Q	I^2	N_{fs}	
All studies combined	198	15641	18037	0.50	0.42	0.57	.00	2032.7*	90.30	297	

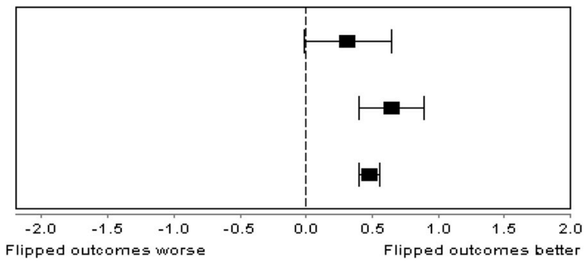
b.	Education level										
	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' g	Lower CI	Upper CI	p	Q	I^2	N_{fs}	
primary	3	121	114	0.47	0.17	0.77	.00	2.60	23.18	4	
secondary	21	902	891	0.64	0.39	0.90	.00	102.03*	80.40	46	
tertiary	174	14618	17032	0.48	0.40	0.56	.00	1885.34*	90.82	244	

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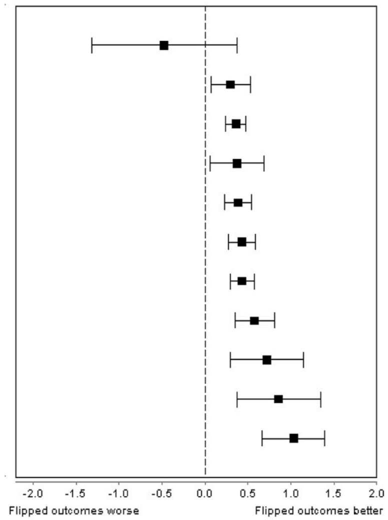
Table 1 (*continued*)

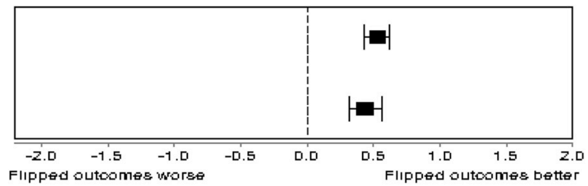
c.	Subject discipline									
	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' g	Lower CI	Upper CI	p	Q	I^2	N_{fs}
library instruction	2	83	71	−0.48	−1.32	0.37	.27	4.80*	79.17	7
IT	14	549	647	0.30	0.07	0.52	.01	52.58*	75.28	7
mathematics	46	3478	4169	0.35	0.23	0.46	.00	228.18*	80.28	35
health sciences	3	75	76	0.35	−0.15	0.84	.17	4.49	55.45	2
business	10	1456	1901	0.38	0.23	0.53	.00	33.07*	72.79	9
medical	50	4117	4686	0.42	0.26	0.57	.00	583.82*	91.61	55
physical sciences	33	3638	3897	0.42	0.28	0.56	.00	240.14*	86.67	36
equine science	1	130	173	0.58	0.35	0.81	.00	–	–	2
engineering	15	841	1244	0.72	0.29	1.15	.00	261.23*	94.64	39
teaching	7	299	205	0.75	0.29	1.20	.00	32.47*	81.52	19
humanities	34	1020	1013	0.98	0.63	1.32	.00	441.64*	92.53	133



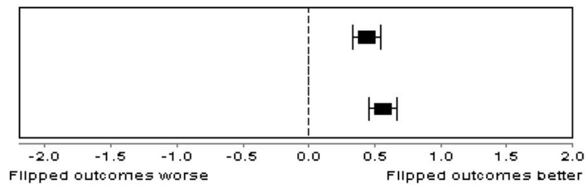
Note: References Table 1:
(a) 1-198.
(b) *primary*: 6, 129, 176; *secondary*: 2, 4, 5, 8, 18, 27, 41, 43; 44; 45, 54, 63, 79, 101, 104, 114, 118, 158, 163, 178, 194; *tertiary*: 1, 3, 7, 9-17, 19-26, 28-40, 42, 46-53, 55-62, 64-78, 80-100, 102, 103, 105-113, 115-117, 119-128, 130-157, 159-162, 164-177, 179-193, 195-198.
(c) *library instruction*: 34, 77; IT: 49, 52, 60, 66, 104, 109, 118, 130, 148, 166, 171, 176, 177, 178; *mathematics*: 13, 15, 27, 29, 33, 35, 39, 41, 44, 53, 54, 62-64, 69, 79-81, 84, 87, 93, 97, 101, 116, 118, 120, 121, 125, 135, 139, 141, 142, 153, 157, 159, 163, 181, 182, 188, 189, 193, 194, 198; *health sciences*: 24, 45, 72; *business*: 7, 16, 22, 25, 42, 106, 113, 160; *medical*: 14, 21, 30, 32, 36, 47, 48, 51, 59, 67, 70, 74-76, 82, 86, 90, 98, 102, 107, 115, 117, 119, 122, 124, 126, 128, 133, 134, 136, 137, 140, 143-145, 149, 151, 154, 167, 169, 172-174, 186, 190; *physical sciences*: 17, 18, 20, 26, 38, 40, 50, 58, 61, 68, 73, 78, 85, 88, 89, 94, 110, 114, 118, 127, 129, 138, 146, 147, 152, 158, 161, 162, 164, 170, 175, 179, 194; *equine science*: 132; *engineering*: 23, 28, 37, 43, 55, 57, 83, 100, 123, 155, 156, 180, 184, 185, 192; *teaching*: 3, 9, 10, 12, 56, 71, 108; *humanities*: 1, 2, 4-6, 8, 11, 19, 31, 46, 65, 91, 92, 95, 96, 99, 103, 105, 111, 112, 118, 129, 131, 150, 165, 168, 183, 187, 191, 194, 195, 196.

Table 2
Summary of effect sizes for subject/course, instructor, pre-class engagement, in-class activity, and group work.

a.	Type of flipped design									
	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' *g*	Lower CI	Upper CI	*p*	*Q*	I^2	N_{fs}
classic	134	10649	12265	0.52	0.43	0.61	.00	1425.9-9*	90.67	214
minimal	64	4992	5772	0.45	0.33	0.58	.00	583.61*	89.21	80



b.	Same instructor used										
	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' *g*	Lower CI	Upper CI	*p*	*Q*	I^2		
N_{fs}	yes	94	7439	8724	0.42	0.32	0.51	.00	708.25*	86-.87	103
no/not reported	104	8202	9313	0.59	0.47	0.71	.00	1312.16*	92.15	203	



c.	Type of pre-class engagement										
	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' *g*	Lower CI	Upper CI	*p*	*Q*	I^2		
N_{fs}	reflec-tive	29	2510	2695	0.35	0.19	0.-51	.00	190.-99*	85-.34	22
not reported	59	4072	4500	0.48	0.33	0.62	.00	543.04*	89.32	83	

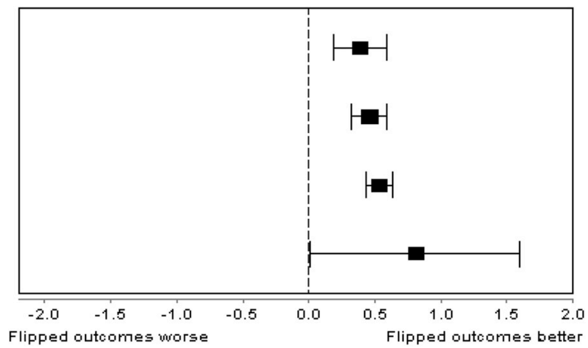


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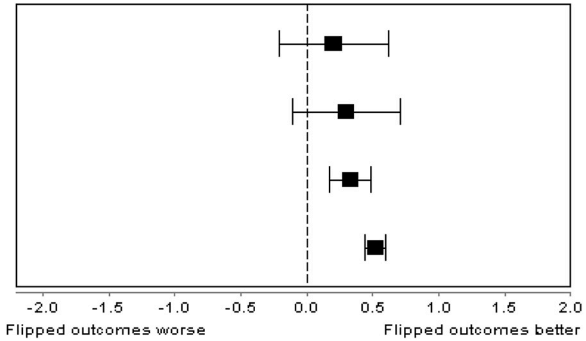
Table 2 (*continued*)

c.	Type of pre-class engagement										
	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' *g*	Lower CI	Upper CI	*p*	*Q*	I^2		
scored	100	8510	10202	0.53	0.43	0.63	.00	1034.2-8*	90.43	165	
discussion	10	549	640	0.78	0.12	1.43	.02	223.37*	95.97	29	
d.	Type of in-class activity										
	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' *g*	Lower CI	Upper CI	*p*	*Q*	I^2		
N_{fs}	tea-cher	4	333	416	0.06	−0.34	0.-46	.33	15.03	80-.04	3
not reported	9	519	477	0.32	−0.15	0.80	.16	85.14*	91.78	5	
standard	2	310	321	0.33	0.17	0.49	.00	0.39	–	1	
student	184	14479	16823	0.52	0.44	0.60	.00	1896.6-8*	90.35	294	
e.	Group work used										
	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' *g*	Lower CI	Upper CI	*p*	*Q*	I^2		
N_{fs}	yes	134	11195	12766	0.46	0.38	0.-55	.00	132-6.09*	89-.44	174
no/not reported	64	4446	5271	0.59	0.44	0.75	.00	685.34*	91.93	125	
f.	Size of in-class groups										
	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' *g*	Lower CI	Upper CI	*p*	*Q*	I^2	N_{fs}	



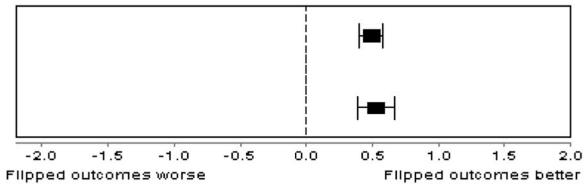


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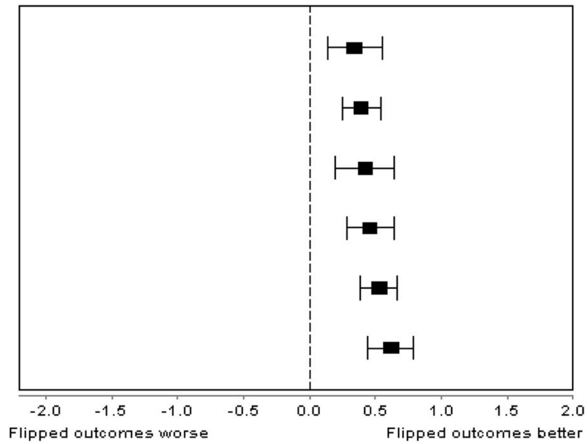
Table 2 (*continued*)

f.	Size of in-class groups									
	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' *g*	Lower CI	Upper CI	*p*	*Q*	I^2	N_{fs}
> 11	4	598	677	0.26	−0.12	0.65	.00	33.52	91.05	1
6–10	7	514	722	0.30	0.09	0.50	.00	17.41*	65.53	4
'small group'	28	2523	2677	0.38	0.22	0.54	.00	199.65*	86.48	25
< 5	52	4136	4481	0.48	0.32	0.64	.00	614.46*	91.70	73
'group'	50	3359	4357	0.57	0.41	0.72	.00	447.11*	89.04	93
not reported	57	4511	5123	0.59	0.44	0.75	.00	694.34*	91.93	111



Note: References Table 2:
(a) *classic*: 1-4, 6, 8-18, 22, 23, 26, 31-34, 37, 39, 40, 43, 46, 47, 50-55, 58, 59, 62-64, 66-72, 76-78, 80, 84, 85, 87, 89, 91-95, 97, 99-102, 104, 106-108, 110-112, 114, 116, 118, 120, 121, 123, 125, 127-140, 142, 144, 148, 150, 151, 152, 155-168, 170-174, 176-180, 182, 185, 186, 188-193, 196, 198; *other*: 5, 7, 19, 20, 21, 24, 25, 27, 28, 29, 30, 35, 36, 38, 41, 42, 44, 45, 48, 49, 56, 57, 60, 61, 65, 73-75, 79, 81-83, 86, 88, 90, 96, 98, 103, 105, 109, 113, 115, 117, 119, 122, 124, 126, 141, 143, 145-147, 149, 153, 154, 169, 175, 181, 183, 184, 187, 194, 195, 197.
(b) *yes*: 3, 7, 8, 10, 11, 14-16, 20, 21, 23-25, 27, 28, 31-34, 36-38, 42, 45, 46, 52-54, 57, 59, 62, 64-66, 68, 69, 71, 77, 79-81, 83, 85, 87, 88, 92, 94, 97, 101, 102, 105, 106, 108, 109, 110, 112, 113, 114, 116, 118-120, 123, 125, 128, 132, 134, 138, 140, 141, 143, 150-152, 156, 161, 162, 164, 166, 167, 169, 171, 175-177, 179, 180, 185, 187, 192, 194, 196-198; ***no***: 1, 2, 4-6, 9, 12, 13, 17, 18, 19, 22, 26, 29, 30, 35, 39, 40, 41, 43, 44, 47-51, 55, 56, 58, 60, 61, 63, 67, 70, 72-76, 78, 82, 84, 86, 89; 90; 91, 93, 95, 96, 98-100, 103, 104, 107, 111, 115, 117, 121, 122, 124, 126, 127, 129-131, 133, 135-137, 139, 142, 144-149, 153-155, 157-160, 163, 165, 168, 170, 172, 173, 174, 178, 181-184, 186, 188-191, 193, 195.
(c): *reflective*: 2, 9, 15, 33, 39, 45, 54, 58, 59, 63, 69, 80, 84, 89, 97, 107, 112, 127, 129, 131, 133, 138, 161, 162, 165, 168, 176, 182, 191, 196; *no/not reported*: 5, 19, 24, 25, 27, 28, 30, 35, 36, 38, 41, 42, 44, 45, 48, 49, 56, 57, 60, 61, 65, 73-75, 79, 81-83, 86, 88, 90, 96, 98, 103, 105, 109, 113, 115, 117, 119, 122, 124, 126, 141, 143, 145; 146; 147, 149, 154, 169, 175, 177, 181, 184, 187, 194, 195, 197; *scored*: 3, 4, 6, 7, 11-14, 16-18, 21-23, 26, 31, 32, 34, 37, 40, 43, 46, 47, 50, 51, 53, 55, 62, 64, 67, 68, 70-72, 76, 78, 85, 87, 91, 93-95, 99-102, 104, 106, 108, 110, 111, 114, 116, 118, 120, 121, 123, 125, 128, 130, 132, 134-136, 139, 140, 142, 144, 148, 150, 151, 152, 153, 155-160, 163, 164, 166, 167, 170-173, 176, 178-180, 183, 185, 186, 188-190, 192, 193, 198; *discussion*: 1, 8, 10, 29, 52, 66, 77, 92, 137, 174.
(d): *teacher*: 29, 106, 168, 197; *not reported*: 20, 35, 41, 44, 82, 124, 153, 154, 183; *standard*: 21, 60; *student*: 1-20, 22-28, 30-34, 36-40, 42, 43, 45-59, 61-81, 83-85, 86, 87-105, 107-123, 125-152, 155-167, 169-182, 184-196, 198.
(e): *yes*: 1-4, 6, 8-14, 17, 20, 22-25, 27-29, 31-34, 36-40, 42, 43, 46, 50, 51, 53, 55, 57-59, 61-64, 66-70, 72, 75-81, 83, 85, 88, 89, 91, 92, 94, 97-100, 102, 105-108, 110-114, 116, 118, 120-123, 125-128, 130, 132, 133, 135, 136, 138, 140-142, 144, 146, 147, 149, 151, 152, 155-157, 159, 161-168, 170-174, 176, 180, 182, 184, 185, 187, 188, 191; 192; 193; 194, 196-198; *not reported*: 5, 7, 15, 16, 18, 19, 21, 26, 30, 35, 41, 44, 45, 47-49, 52, 54, 56, 60, 65, 71, 73, 74, 82, 84, 86, 87, 90, 93, 95, 96, 101, 103, 104, 109, 115, 117, 119, 124, 129, 131, 134, 137, 139, 143, 145, 148, 150, 153, 154, 158, 160, 169, 175, 177; 178; 179, 181, 183, 186, 189, 190, 195.
(f):*6-10*: 32, 69, 76, 106, 136, 149, 197; *<5*: 8, 11, 13, 23, 38, 39, 42, 43, 46, 50, 57, 58, 61, 62, 72, 77, 83, 85, 89, 91, 92, 95, 97, 99, 101, 102, 103, 108, 111, 120, 122, 123, 126, 132, 133, 135, 137, 141, 146, 147, 152, 155, 156, 162, 163, 164, 171, 173, 174, 176, 184, 185; *>11*: 29, 70, 98, 151; *'small group'*: 10, 12, 20, 22, 24, 25, 36, 55, 66, 78, 88, 100, 105, 113, 118, 121, 125, 128, 138, 140, 144, 157, 159, 172, 182, 187, 188, 193; *not reported*: 4, 5, 7, 15, 18, 19, 21, 26, 30, 31, 35, 40, 41, 44, 45, 47-49, 52, 59, 60, 71, 73, 74, 82, 84, 86, 90, 93, 96, 104, 109, 115, 117, 119, 124, 129, 131, 134, 139, 143, 145, 148, 150, 153, 154, 158, 175, 177-179, 181, 183, 186, 189, 190, 195; *'group'*: 1, 2, 3, 6, 9, 14, 16, 17, 27, 28, 33, 34, 37, 51, 53, 54, 56, 63, 64, 65, 67, 68, 75, 79, 80, 81, 87, 94, 107, 110, 112, 114, 116, 127, 130, 142, 160, 161, 165-168, 170, 180, 191, 192, 194, 196, 198.

individual studies (Borenstein, Hedges, Higgins, & Rothstein, 2010). I^2 was used to measure the proportion of variance in the individual study effect sizes that was not attributable to sampling error, with I^2 values of 25%, 50% and 75% indicating low, moderate and high levels (respectively) of heterogeneity (Higgins, Thompson, Deeks, & Altman, 2003). Heterogeneity was further addressed by the use of sub-group analyses, which examined whether effect sizes obtained from different studies varied according to a number of methodological and sampling variables (e.g., subject discipline, type of pre-class activity).

Finally, Orwin's (1983) Fail Safe *N* (N_{fs}) statistics were calculated in order to assess the potential bias caused by the tendency for journals to favor statistically significant results (publication bias) (Rosenthal, 1995) and the fact that unpublished/non peer-reviewed papers (gray literature) were excluded from our analyses. The *N*fs estimates the number of unpublished studies needed to reduce a finding to a small/trivial effect (defined here as $g = .2$) and the larger the *N*fs, the more confident we can be in a finding (Borenstein et al., 2009).

3. Results

We summarize the results of the meta-analysis in four tables. Table 1 reports effect sizes for the overall effect, and for moderating effects of education level (primary, secondary, tertiary) and subject discipline (business, engineering, equine science, health sciences, humanities, IT, library instruction, mathematics, medical, physical sciences, teaching). Table 2 reports moderating effects of the flipped classroom itself, specifically, the 'type' of flipped design that was used (classic, other); whether the instructor was the same across flipped and traditional approaches (yes, no/not reported); type of measure used for pre-class engagement (scored, discussion, reflective, not reported); type of in-class activity (student-guided, teacher-guided, standard didactic, not reported); whether there were group activities (yes, not reported); and, if so, the size of the group (< 5, 6–10, > 11, small group, group, not reported). Table 3 reports moderating effects of type of assessment (course, exam, skill, post-test), and quality/volume of assessment (robust, less robust, not reported). Table 4 reports moderating effects of whether means and SDs were estimated or not; and whether the study design was between- or within-subjects. All effect sizes discussed in the text are significant (i.e., $p < .05$) unless otherwise indicated.

3.1. Overall effect of flipped classrooms vs traditional learning approaches

As indicated in Table 1a, across all studies and irrespective of moderators, flipped classrooms exerted a moderate effect on student performance relative to traditional learning approaches (Hedge's $g = 0.50$). Orwin's failsafe N was high (Nfs = 297) indicating that we can be confident in this finding.

3.2. Moderating effect of education level

As indicated in Table 1b, flipped classrooms were associated with better student performance across all levels of education. The effect was moderately strong among secondary students ($g = 0.64$), and moderate among tertiary students ($g = 0.48$) and primary school students ($g = 0.47$), although the latter is based on only three studies.

3.3. Moderating effect of discipline

Table 1c summarizes effect size according to discipline. The flipped classroom tended to be beneficial regardless of discipline. The strongest effect sizes occurred in the disciplines of humanities ($g = 0.98$; $k = 34$), teaching ($g = 0.75$; $k = 7$), and engineering ($g = 0.72$; $k = 15$). There were weaker effect sizes for IT ($g = 0.30$; $k = 14$) and mathematics ($g = 0.35$; $k = 46$) and business ($g = 0.38$, $k = 10$). Table 1c also shows that there were only two non-significant effects, that for library instruction ($g = -0.48$, $p = 0.27$) and health sciences ($g = 0.35$, $p = 0.17$), although these results are based on only two and three studies, respectively. Elsewhere, N_{fs} values varied according to discipline (range N_{fs}: 2–133) and the lower values do, however, indicate that more research is needed in some disciplines in order to increase the confidence in these findings.

3.4. Moderating effect of type of flipped design

Table 2a shows that the flipped effect is somewhat larger when the study used a classic flipped design (i.e., pre-class preparation of any type, some form of pre-class engagement measure, and student activities in class with or without in-class lecturing) ($g = 0.52$) compared to when the study did not incorporate all three of these elements (i.e., minimally flipped; $g = 0.45$).

3.5. Moderating effect of instructor

As shown in Table 2b, flipped classrooms were likely to have a weaker effect on student performance if the instructor was the same across both flipped and traditional modes of teaching ($g = 0.42$), compared with when the instructor differed between modes or the information was not reported ($g = 0.59$).

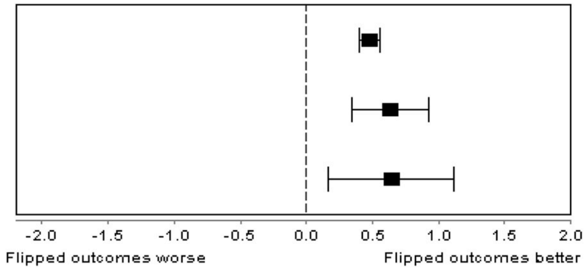
3.6. Moderating effect of type of pre-class engagement

We classified pre-class engagement as not reported ($k = 59$), discussion ($k = 10$), reflective ($k = 29$) or scored in some way, e.g. a

Table 3
Summary of effect sizes for assessment.

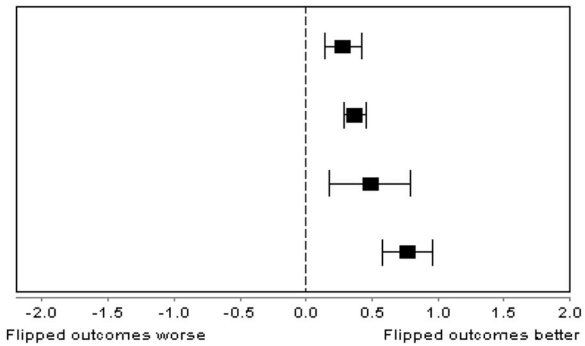
a. Quality and volume of assessment

	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' *g*	Lower CI	Upper CI	*p*	*Q*	I^2	N_{fs}
robust	171	14237	16790	0.47	0.39	0.54	.00	1534.57*	88.92	231
not reported	13	621	499	0.76	0.40	1.12	.00	92.46*	87.02	36
less robust	14	783	748	0.76	0.15	1.36	.01	363.20*	96.42	39



b. Type of assessment

	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' *g*	Lower CI	Upper CI	*p*	*Q*	I^2	N_{fs}
course	26	2241	2426	0.28	0.14	0.42	.00	117.47*	78.72	10
exam	106	10923	12780	0.37	0.29	0.46	.00	979.45*	89.28	90
skill	10	619	610	0.49	0.18	0.79	.00	49.02*	81.64	15
post-test	70	2911	3276	0.77	0.58	0.96	.00	816.38*	91.55	200

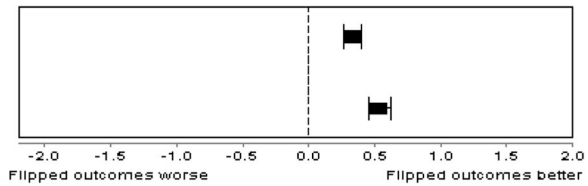


Note: References Table 3:
(a) *robust*: 1, 3-26, 28-32, 35, 38-43, 46-48, 50-53, 55, 57-62, 64, 66-70, 71, 72, 74-76, 78-100, 102, 103, 105-115, 117-123, 125-142, 144-148, 15, 152, 154-173, 175, 177-189, 191, 192, 193, 195, 197, 198; *not reported*: 37, 44, 45, 49, 54, 56, 73, 104, 116, 150, 153, 174, 176, 194; *less robust*: 2, 27, 33, 34, 36, 63, 65, 77, 101, 124, 143, 149, 190, 196.
(b) *course*: 16, 28, 40, 53, 56, 58, 61, 64, 79, 83, 84, 109, 113, 116, 130, 146, 156, 165, 166, 173, 176, 178, 186, 187, 188; *exam*: 3, 7, 14; 13; 15, 19, 20, 22-26, 29, 30, 33, 35, 36, 38, 39, 42, 50, 53, 55, 57, 59, 62, 67, 68, 69, 70, 71, 72, 74, 75, 76, 78, 80-82, 84, 85-90, 93, 94, 97, 99, 102, 106, 107, 108, 110, 111; 112; 113, 115, 117, 118, 119, 120, 121, 123, 125, 126, 127, 128, 132, 133, 134, 135, 138, 139, 140, 141, 142, 143, 144, 145, 147, 150, 151, 152, 154, 155, 157, 158, 166, 167, 170, 171, 172, 173, 179, 181, 182, 185, 187, 188, 190, 192, 195, 197, 198; *skill*: 21, 48, 51, 52, 66, 74, 90, 100, 137, 177; *post-test*: 1, 2, 4, 5, 6, 8, 9, 10, 11, 12, 17, 18, 27, 31, 32, 34, 37, 41, 43; 44; 46; 45; 47, 49, 54, 60, 63, 65, 73, 77, 80, 88, 91, 92, 96, 98, 101, 103, 104, 105, 111, 114, 122, 124, 129, 131, 136, 138, 148, 149, 153, 159-164, 168, 169, 174, 175, 180, 183, 184, 189, 191, 193, 194, 196.

Table 4
Summary of effect sizes for study information.

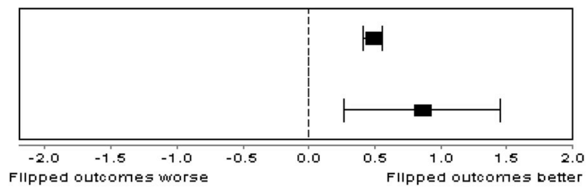
a. Estimated M/SD data

	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' g	Lower CI	Upper CI	p	Q	I^2	N_{fs}
yes	17	1314	1990	0.33	0.26	0.40	.00	16.23	1.39	11
no	181	14327	16047	0.53	0.45	0.62	.00	2006.60*	91.03	299



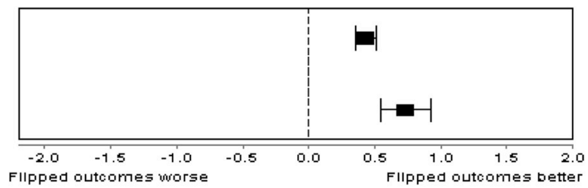
b. Study design

	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' g	Lower CI	Upper CI	p	Q	I^2	N_{fs}
between	190	15274	17671	0.49	0.41	0.56	.00	1907.63*	90.09	276
within	8	367	366	0.86	0.26	1.45	.00	96.88*	92.77	26



c. Extent of the flipped subject/course

	$N_{studies}$	$N_{flipped}$	$N_{traditional}$	Hedges' g	Lower CI	Upper CI	p	Q	I^2	N_{fs}
entire	148	12162	14506	0.42	0.34	0.50	.00	1292.97*	88.63	163
part	50	3479	3531	0.77	0.56	0.94	.00	688.90*	92.89	143



Note: References Table 4:
(a) *yes*: 21, 25, 28, 32, 35, 39, 45, 62, 72, 76, 89, 106, 122, 125, 150, 160, 166; *no*: 1-20, 22-24, 26, 27, 29-31, 33, 34, 36-38, 40-44, 46-61, 63-71, 73-75, 77-88, 90-105, 107-121, 123, 124, 126-149, 151-159, 161-165, 167-198.
(b) *between*: 1-36, 38-45, 47-85, 87-130, 132-139, 141-148, 150-152, 154-190, 192-198; *within*: 37, 46, 86, 131, 140, 149, 153, 191.
(c) *entire*: 1, 3-10, 14-16, 19, 20, 22-26, 28, 29, 31, 32, 34, 35, 38, 40-42, 44, 48-50, 52-69, 72, 74-85, 87-99, 103-109, 111, 112, 113-118, 120, 121, 123, 124, 126-128, 130, 132, 134, 135, 138, 139, 141, 142, 144, 146-148, 151, 152, 155-160, 162, 163, 165-171, 173, 175-178, 180-182, 184-189, 192, 193, 195-198, *part*: 2, 11-13, 17, 18, 21, 27, 30, 33, 36, 37, 39, 43, 45-47, 51, 70, 71, 73, 86, 100-102, 110, 119, 122, 125, 129, 131, 133, 136, 137, 140, 143, 145, 149; 150, 153, 154, 161, 164, 172, 174, 179, 183, 190, 191, 194.

quiz ($k = 100$). The flipped classroom had the strongest effect on student performance when students engaged in pre-class discussion ($g = 0.78$), and the weakest effect when students reflected on pre-class material ($g = 0.35$)—notably, slightly weaker than when there was pre-class engagement was not reported ($g = 0.48$). There was a moderate effect size when responses to pre-class materials were scored in some way ($g = 0.53$) (see Table 2c).

3.7. Moderating effect of in-class activity

Once in-class, most studies reported a student-centred approach ($k = 184$), which yielded a moderate effect size ($g = 0.52$). The effect of teacher-centred ($k = 4$) approaches was non-significant ($g = 0.06$, $p = 0.75$). This was also the case when the type of in-class activity was not reported ($g = 0.32$, $p = 0.19$). Although there was a significant effect for a standard didactic lecture during the in-class session ($g = 0.33$), this result is based on only two studies (see Table 2d). However, other than student-centred in-class activities ($N_{fs} = 294$), the associated N_{fs} statistics for the remaining categories were consistently low (≤ 5), hence these data should be treated cautiously.

3.8. Moderating effect of group size

There was minimal difference in effects when we coded studies according to whether authors specifically reported that there was group work ($g = 0.46$) or not ($g = 0.59$) (see Table 2e). When we were able to code more specifically for group numbers, effects ranged from 0.26 (groups greater than 11) to 0.30 (groups of 6–10) and 0.48 (groups of 2–5). Where we could only code more generally, effects ranged from 0.38 ('small group') to 0.57 (group size not indicated) (see Table 2f).

3.9. Moderating effect of type of assessment

We tested the extent to which the quality of student performance measures moderated flipped classroom effects. We classified the vast majority of assessments ($k = 171$) as 'robust', and these yielded a moderate effect ($g = 0.47$). Less robust measures ($k = 14$) were associated with stronger effects ($g = 0.76$), as were those studies where there was insufficient detail provided about the nature of assessment ($k = 13$; $g = 0.76$) (see Table 3a). We then undertook a more fine-grained analysis, coding measures of student performance according to whether the measure was the overall course mark; an exam mark; a post-test mark associated with a specific learning activity; or some skill-based measure. Flipped classrooms had the strongest effect when student performance was measured on the basis of an immediate post-test ($g = 0.77$) and was weakest when the measure was final course mark or grade ($g = 0.28$) or exam ($g = 0.37$). There was a moderate effect for tests of skill ($g = 0.49$) (Table 3b).

3.10. Moderating effect of study design

Table 4a shows that the flipped classroom effect was stronger when actual means and SDs were used ($g = 0.53$) compared to when they were estimated ($g = 0.33$). In addition, flipped classrooms had a stronger effect when the study design was within-participants ($g = 0.86$) compared to when it was between-subjects ($g = 0.49$) (Table 4b). Finally, the effect was greater when only part of a course/subject was flipped ($g = 0.77$) compared to the entire course/subject ($g = 0.42$) (Table 4c).

4. Discussion

Overall, flipping a classroom has a positive, moderate effect on student performance. Specifically, the moderate effect reflects half a standard deviation improvement in performance relative to traditional teaching models—which, in practical terms, is substantial. Context is important, of course, but for many students an average increase in scores of half a standard deviation could represent a shift to a higher-grade category. In Hattie's (2017) analysis of influences on student outcomes, an effect size of 0.50 is comparable to that associated with strategies emphasising feedback such as questioning and formative evaluation as well as the use of technologies such as intelligent tutoring systems, computer assisted instruction, and the use of mobile phones. It is twice as large as the effect sizes due to class size, teacher personality, programmed, discovery based or individualised instruction, and technologies such as clickers and one-on-one laptops. The positive moderate effect is somewhat stronger than those reported in previous smaller, localised meta-analyses in health professions education ($es = 0.39$, $k = 28$; Hew & Lo, 2018) and mathematics ($es = 0.30$, $k = 21$; Lo et al., 2017); substantially smaller than those observed for nursing knowledge ($es = 1.06$) and skill ($es = 1.40$) ($k = 12$; Hu et al., 2018); and stands in stark contrast to the six pharmacy education studies meta-analysed by Gillette et al. (2018) where no significant effect was found.

The effect in our meta-analysis is magnified among secondary students ($g = 0.64$; $k = 21$) whereas in tertiary settings it effectively remains unchanged ($g = 0.48$; $k = 174$). We speculate that the difference may be due to the greater control that secondary teachers have over their students in terms of mandating and monitoring homework, for example; and the fact that secondary teachers are literally in the classroom every (or most) day(s) and therefore have greater opportunity for personalized interaction with each of their students. Abeysekera and Dawson (2015) examined the theoretical underpinnings of flipped classrooms. Based on theories of self-determination and cognitive load they proposed that a flipped learning environment could encourage greater levels of intrinsic and extrinsic motivation through the satisfaction of students needs for relatedness (to the instructor and each other), autonomy, and competence. It is possible that such outcomes would show up to greater effect in environments where students may not typically have experienced them, such as secondary school.

There is notable disparity when we examine the results according to discipline, although one should be cautious about interpreting effect sizes that are based upon small numbers of studies. There were relatively weak effects for health sciences (note that $k = 3$), IT, and mathematics (gs = 0.30 to 0.35), weak-moderate effects for business, medical, and physical sciences (gs range from 0.38 to 0.42); a moderate effect for equine science ($g = 0.58$, albeit $k = 1$); moderately strong effects for engineering ($g = 0.72$) and teaching ($g = 0.75$, $k = 7$), and a strong effect in humanities ($g = 0.98$). It is difficult to discern a pattern that might explain these divergent findings. However, one might tentatively speculate that there is a tendency for effect sizes to be smaller in those disciplines that are arguably more amenable to problem-solving activities that yield clearly defined solutions (e.g., in mathematics, students work through a problem that has a right or wrong answer). It is possible that traditional courses in these disciplines—e.g., mathematics, IT, business, medicine—are structured in such a way that students are already provided with scaffolding, so that flipping a classroom does not add as much value as in disciplines where active learning and problem-solving is less prevalent or normative (e.g., humanities). In short, it may be that the flipped classroom effect is not as strong in those disciplines that are already highly structured when it comes to student learning (e.g., working through a problem in mathematics versus discussing art history or politics in the humanities). Alternatively, it could be that the nature of the assessment used in Humanities shows the effects of prepared, active engagement with ideas catalysed by a flipped classroom better than those in the sciences.

Abeysekera and Dawson (2015) postulated that the self-paced nature of pre-class activities as well as facilitating tailored instruction may reduce cognitive load and thus improve learning outcomes. We might expect, then, to see the effects of flipping even when all aspects of the method are not implemented. Indeed, there was little difference in effect sizes when the classroom was classified as being 'classically' flipped ($g = 0.52$) compared to when it was minimally flipped ($g = 0.45$). Relatedly, there was a negligible difference between flipped classrooms that scored pre-class engagement exercises (e.g., quizzes; $g = 0.53$) and those where there was no reported measure of pre-class engagement ($g = 0.48$). Indeed, *not* engaging pre-class yielded a slightly stronger effect on student performance compared to when students reflected on pre-class materials ($g = 0.35$) (although there was a strong effect of 0.78 when students engaged in discussion about pre-class materials, we are cautious about over-interpreting this result because it is based on only 10 studies). Together, these findings suggest that that even a rudimentary approach to flipping—that is, ensuring pre-class and within-class engagement without any pre-class testing—is sufficient to attain a moderate improvement in student performance. Perhaps this outcome should be unsurprising. Flipped classrooms require substantial organisation and commitment prior to class from student and teacher. As such, it may be that one important contributor to enhanced learning processes in the flipped classroom is the greater effort that educators invest when clarifying to students the nature of tasks and the intended learning outcomes (e.g., Bergmann & Sams, 2014). That being the case, pre-testing of that learning may have only incidental impact, especially as flipped classes often embed directly applicable pre-learning in in-class activities.

Once in class, the vast majority of studies reported that the in-class activities were student-centred ($k = 184$), so that the overall effect size barely changed ($g = 0.52$). Relatedly, it does not seem to matter too much whether students engage in group work ($g = 0.46$) or work individually ($g = 0.59$); the most important thing is that in-class activities are student-centred. In addition, when students do work in groups there was little evidence that group size matters.

Perhaps not surprisingly, the flipped effect was somewhat stronger when the instructor differed between traditional and flipped modes ($g = 0.59$) compared to when the instructor was the same ($g = 0.42$). We interpret this finding as reflecting the role of teacher personality in helping to motivate students to perform better (e.g., Darling-Hammond, 2000). It is possible that teachers willing to try new ways to engage students tend to be those who are already engaging, so that it is their enthusiasm for the flipped classroom model that helps to enhance its effect.

Finally, the flipped effect was strongest when student performance was measured immediately following a flipped session (g = 0.77) compared to when student performance was a final course mark or grade (g = 0.28). These findings are to be expected. As a robust literature on retention indicates, students perform better when they are tested immediately on material they have just learned, but the effect fades over time (e.g. Roediger & Karpicke, 2006; Ferreira, Maguta, Chissaca, Jussa, & Abudo, 2016; van Vliet, Winnips, & Brouwer, 2015). In addition, an overall mark or grade does not necessarily reflect assessment of flipped learning per se. That said, we think it is notable that [a] the active learning activities that characterize the flipped classroom lead to a strong immediate effect on performance compared with when performance is measured immediately in a traditional teaching approach and [b] even when an overall mark or grade is used, the flipped classroom is still associated with an average improvement of around 1/3 of a standard deviation.

4.1. Limitations and future research

The major limitation of meta-analysis is that conclusions are contingent upon the quality of the research that was conducted in the first place (e.g., Cooper & Hedges, 2009). Although it was beyond the scope of our review to code for studies that explicitly tested a flipped vs traditional approach using experimental design in the laboratory, we did not come across any such study during the initial, dynamic process of deciding on appropriate categories for coding. Thus, most likely the studies reported here would be classified as field studies and therefore are inherently 'noisy'. For example, logistical and practical considerations would make it difficult for a teacher to counter-balance conditions if he or she was employing a within-subjects experimental design, and therefore we cannot discount order effects in such a design. Similarly, a teacher may employ a between-subjects design, but often such an approach means comparing two cohorts from two different years (e.g., one exposed to traditional teaching, the other to a flipped classroom), so that there is a confounding effect of cohort.

A further limitation of these studies is they typically would not have been able to control for pre-existing student ability. That said, the comprehensive nature of this meta-analysis—i.e., it features results from almost 34,000 students across 198 studies—is such that methodological limitations in individual studies are diluted. Thus, the moderate flipped effect size that we have obtained here is likely a fairly true reflection of the state of play.

Future researchers could therefore explicitly test the idea the flipped effect can be primarily attributed to active learning. For example, while there are certainly logistical and practical benefits to pre-recording fundamental information for students to engage with in their own time, a traditional lecture linked to a tutorial that involves active learning opportunities should be just as effective as the flipped model (that said, if lecture material was pre-recorded, then the time previously spent in the lecture theatre could also be used for more interactive activities). It is also notable that our results suggest that group size during the in-class activities does not matter. Although we did not code for activities where $n = 1$, future researchers could test the extent to which individuals can effectively engage in active learning exercises on their own as part of a flipped course. Such an approach would be especially amenable given the increased emphasis on fully online learning in universities.

Finally, and more broadly, we note that the vast majority of studies have been conducted at the tertiary level ($n = 174$), with comparatively few ($n = 21$) conducted with secondary students, and barely any with primary school students ($n = 3$). A plausible explanation for this is that many tertiary educators are also researchers and can thus carry out research as part of their normal role. We think the disparity is notable, if only because the empirical evidence for flipped classrooms is not commensurate with the push to implement flipped classrooms at secondary school level. Thus, there is scope for future researchers to expand our understanding of the effects of the flipped classroom at primary and secondary levels. In so doing, researchers could further examine the potential moderating effects we have tested here. For example, as student learning becomes more specialized with age, might one size fit all at primary level, but become increasingly discipline-specific by the time students reach tertiary education?

4.2. Conclusion

This meta-analysis of 198 studies from a wide variety of disciplines and encompassing predominantly tertiary but also secondary school and primary levels, and featuring data from almost 34,000 students, revealed that the flipped classroom has a moderate positive effect on student performance (specifically, $g = 0.50$)—an effect that, in practical terms, is meaningful for students.

We therefore conclude that flipping the classroom is likely to lead to an improvement in student performance, to varying degrees depending on discipline. We think the next key questions for future researchers to answer are, *what* is the important ingredient in the flipped classroom recipe and *how* does it contribute to the effectiveness of the flipped classroom? As others have already suggested (Abeysekera & Dawson, 2015), what distinguishes a flipped classroom from traditional teaching is not so much that information is pre-recorded, or that students engage with fundamental information using technology outside the lecture theatre but, rather, that students have an opportunity to engage in active learning and problem-solving with hands-on guidance provided by an expert. Moreover, the structure of the flipped classroom model provides more time for students to engage in active learning practices, thereby further increasing the effectiveness of these practices.

Our results are consistent with this perspective. We have found that the flipped effect is strongest in disciplines that arguably have been (traditionally) less structured and less likely to employ active learning approaches; that it is stronger at secondary level where teachers have more opportunity for personalized engagement; that even the most basic approach to flipping—i.e., providing pre-recorded material and engaging with students in-class—is associated with a moderate improvement in student performance; and that it is not group work per se that matters but, rather, that in-class activities are student-centred (as opposed to being didactic). In short, we speculate that the flipped classroom is effective because it encourages teachers to be more structured and better-organized; to think through the practice-oriented implications of the information that they impart; and to therefore construct exercises and tasks for students that better enable students to learn by doing. Put another way, the opportunity for well-planned active learning is probably the key ingredient in the flipped classroom formula (e.g, Freeman et al., 2014).

Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.edurev.2020.100314.

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Subscript numbering 1–198 refers to studies that are included in the meta-analysis; numbers are cross-referenced to studies listed in Tables 1–4

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